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Paper Code : PCC-CSBS 502 Database Management Systems

UPID : 005963

Time Allotted : 3 Hours

Full Marks :70

The Figures in the margin indicate full marks.

Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer *any ten* of the following :

[1 x 10 = 10]

- (I) Isolation means _____.
 - (a) transaction must not interfere with each other
 - (b) transaction must interfere with each other
 - (c) transaction must be in a consistent state
 - (d). transaction must be executed immediately
- (II) In distributed database environment, the execution of set of distributed transactions is said to be serializable if an only if
 - (a) the execution of the set of transactions at each site is serializable
 - (b) the serialization orders of set of transactions at all the sites are identical
 - (c) the set of transactions that are not in serial order
 - (d) the serialization orders of set of transactions at all the sites are not identical
- (III) Network-based IDS can be used to monitor —
 - (a) Web server logs
 - (b) When something weird occurs
 - (c) All connections to the database server
 - (d) None of the above
- (IV) Which one of the following given statements possibly contains the error?
 - (a) select * from emp where empid = 10003;
 - (b) select empid from emp where empid = 10006;
 - (c) select empid from emp;
 - (d) select empid where empid = 1009 and Lastname = 'GELLER';
- (V) Why data fragmentation and distribution is considered as a desirable property?
 - (a) Reduced data transmission cost
 - (b) Reduces table size
 - (c) Availability
 - (d) All of the above.
- (VI) If a transaction acquires an exclusive lock, then it can perform ____ operation.
 - (a) read
 - (b) write
 - (c) read and write
 - (d) update
- (VII) Object-oriented database is an alternative implementation of _____model.
 - A. Logical
 - B. Cross
 - C. Relational
 - D. Linear
- (VIII) Each ____ consists of a set of pointers containing the address of the disk block containing the value for that _____.
 - (a) Dataset
 - (b) Data Item
 - (c) Key
 - (d) Log
- (IX) Each search key value in the data file has an index record in the ____ index.
 - (a) Sparse
 - (b) Dense
 - (c) Both A and B
 - (d) None of the above

- (X) Which query is used for sorting data that retrieves the all the fields from empinfo table and listed them in the ascending order?

A) SELECT * FROM empinfo ORDER BY age;
B) SELECT * FROM empinfo ORDER age;
C) SELECT * FROM empinfo ORDER BY COLUMN age;
D) SELECT * FROM empinfo SORT BY age;

- (XI) There are three table above tables A, B and C as given below

Table A

Id Name Age

12 Arun 60
15 Shreya 24
99 Rohit 11

Table B

Id Name Age

15 Shreya 24
25 Hari 40
98 Rohit 20
99 Rohit 1

Table C

Id Phone Area

10 2200 02
99 2100 01

Then determine the number of tuples return in the result of the following SQL query.

```
SELECT A.id  
FROM A  
WHERE A.age > ALL (SELECT B.age FROM B  
WHERE B. name = "arun");
```

- a. 4
b. 3
c. 0
d. 1

- (XII) How can we prevent SQL Injection attack?

- (a) We should pre-define the input type, input field and length of the user data to validate the input for the user authentication.
(b) Access privileges should be restricted for the users
(c) Administrator accounts should not be used.
(d) All of the above

Group-B (Short Answer Type Question)

Answer *any three* of the following :

[5 x 3 = 15]

2. Discuss the different levels of views. [5]
3. Discuss the ACID properties of a transaction. [5]
4. Define query Optimization in brief. [5]
5. Explain the two phase locking with a suitable diagram. [5]
6. Explain the advantages of database management system over file management system. [5]

Group-C (Long Answer Type Question)

Answer *any three* of the following :

[15 x 3 = 45]

7. (a) Prove or disprove: $AB \rightarrow C, A \rightarrow D, CD \rightarrow EF$ then $AB \rightarrow F$ [5]
(b) If A is the set of attributes then write the algorithm to compute A^+ (Closure of attribute set). [5]
(c) Define MVD. Explain the lossless(or non-additive) join property of decomposition. [5]
8. Consider the relational database: Employee (person-name, street, city) Works (person-name, company name, salary) Company (company name, city) Manages (person-name, manager-name) Write down appropriate relational algebra and SQL statement for any three queries:

- (a) Find the name of all employees who work for 'SBI bank'. [5]
- (b) Find name, street address, cities of residence of all employees who work for 'UBI Bank' and earn more than Rs. 5,00,000 per annum. [5]
- (c) Find the second highest salary for employees in 'SBI bank'. [5]
9. (a) Draw an ER diagram for a travel agency consisting of the following: [6]
Customers, buses, drivers, conductors, guides, tickets, booking, agents, reservations, conducted tours and hotels.
- (b) Describe entities, attributes, relationships and primary keys. [4]
- (c) Reduce the ER diagram into relational schema by defining all the constraints and assumptions. [5]
10. (a) Consider insertion sequence: 85, 1, 7, 3, 12, 9, 6, 20, 13. Construct B+ tree. [5]
- (b) Consider the file with $r=30000$ records (fixed-length) of size $R=100$ bytes stored on a disk with block size $B=1024$ bytes. Suppose each index entry in each index file takes 15 (9 bytes for index value and 5 bytes for pointer) bytes. Calculate the number of accessing blocks for clustering index. [7]
- (c) Explain partitioned hashing, its working mechanism and its limitations. [3]
11. (a) Describe ACID properties of a transaction. [4]
- (b) In a concurrent schedule, when do two instructions conflict? How can they be converted to a serializable schedule. Illustrate with an example? [7]
- (c) Differentiate between cascading schedule and a cascadeless schedule. [4]

*** END OF PAPER ***